INDUSTRY TREND ANALYSIS

AI infrastructure becomes a full-stack constraint

Structural · Evidence current to 27 August 2026

The signal

Competition has moved beyond GPU access to power, networks, data centers, financing, cooling, grid access, and asset utilization.

485 to 950 TWh	42 GW	84 projects / RMB 500bn+
IEA central projection for global data-center electricity use from 2025 to 2030. [1]	server load that could be idle by 2030 if power and deployment bottlenecks persist, as cited in the TMT Daily archive. [2]	data-center projects and announced investment cited for Ulanqab; a 1 GW site was estimated to save about RMB 5bn annually from lower power cost. [3]

What is changing - and why it matters

AI infrastructure is becoming a coordinated system of compute, power, cooling, networks, data-center delivery and financing. A bottleneck in any layer can strand capacity elsewhere. The strategic shift is from buying chips to orchestrating a portfolio of long-lead assets and contracts while preserving flexibility as models and hardware change. The evidence supports a structural constraint, but the winning architecture remains unsettled.

Executive implication

AI infrastructure is becoming a coordinated system of compute, power, cooling, networks, data-center delivery and financing. A bottleneck in any layer can strand capacity elsewhere. The strategic shift is from buying chips to orchestrating a portfolio of long-lead assets and contracts while preserving flexibility as models and hardware change. The evidence supports a structural constraint, but the winning architecture remains unsettled.

Action agenda

- Build one capacity plan across compute, power, networks, cooling and financing.
- Contract for flexibility in utilization, hardware refresh and power delivery.
- Use location and energy strategy as product and margin decisions, not facilities decisions.

What could change the view

- Installed chips outrun available power or interconnection.
- High-density designs create stranded equipment or retrofit costs.

Selected evidence

[1] Authoritative research · IEA · 2026-04-16 · Key Questions on Energy and AI · p. 10

[2] Weekly Brief · 2026-08-18 · AI□□□□□□□□□□GPU□□□□□□□□□□□

[3] Weekly Brief · 2026-08-14 · AI□□□□□□□□□□□

[4] Weekly Brief · 2026-05-06 · When AI and Cost Become One Agenda

[5] Weekly Brief · 2026-07-28 · The AI Bill Lands on the CEO’s Desk

[6] Weekly Brief · 2026-08-10 · AI□□□□□□□□□□□□□□□□□□□□

[7] Weekly Brief · 2026-08-26 · AI□□□Agent□□□□□□□□□□□□□□□□□

[8] Authoritative research · ITU · 2025-11-27 · Global Connectivity Report 2025

[9] Authoritative research · Goldman Sachs Research · 2026-08-19 · AI/Data Centers: Power Demand, Cyclical Progression, and Sustainability Implications

[10] Authoritative research · BCG · 2026-08-18 · Powering AI Through Grid-Positive Data Centers

This independent analysis connects the available Weekly Brief archive with selected authoritative research. Forecasts and company-reported figures are identified in context. The complete evidence list remains available on the website.